ANALYSIS I — SHEET 7 (vacation sheet)
Power Series. Radius of Convergence.

1. Find the radius of convergence R of each of the following power series. For each part determine whether or not the series converges at $x = \pm R$. [*Hint: Stirling's formula should prove helpful for (vi).*]

$$\text{(i)} \quad \sum n^{2026} x^n; \qquad \text{(ii)} \quad \sum \frac{x^n}{2^n n^4}; \qquad \text{(iii)} \quad \sum 2^n x^{n!};$$

$$\text{(iv)} \quad \sum \sinh(n) x^n; \qquad \text{(v)} \quad \sum \sin(n) x^n; \qquad \text{(vi)} \quad \sum \frac{(nx)^n}{n!}.$$

2. Define

$$S(z) = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)!}, \qquad C(z) = \sum_{k=0}^{\infty} \frac{z^{2k}}{(2k)!}.$$

(i) Show that the series $S(z)$ and $C(z)$ converge for all complex numbers z.
(ii) Show that $S'(z) = C(z)$ and $C'(z) = S(z)$.
(iii) By differentiating, show that $C(z)^2 = 1 + S(z)^2$.
(iv) By using the Cauchy product for series, show that $S(2z) = 2S(z)C(z)$.

3. What are the sums of the following series for $|x| < 1$?

$$\text{(i)} \quad \sum_{n=0}^{\infty} x^n; \qquad \text{(ii)} \quad \sum_{n=1}^{\infty} n x^n; \qquad \text{(iii)} \quad \sum_{n=1}^{\infty} n^2 x^n.$$

4. Find the radius of convergence of the power series

$$J_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{x}{2}\right)^{2n}.$$

Assuming term-by-term differentiation is allowed within the disc of convergence show that $y = J_0(x)$ is a solution of the equation

$$xy'' + y' + xy = 0.$$

5. The *Fibonacci numbers* F_n are defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+2} = F_{n+1} + F_n$ for $n \geqslant 0$. Define

$$F(x) = \sum_{n=0}^{\infty} F_n x^n.$$

Determine the radius of convergence of $F(x)$. By summing the identity $F_{n+2}x^n = F_{n+1}x^n + F_n x^n$ from $n = 0$ to ∞, or otherwise, find $F(x)$ in closed form.

6. (i) Let p_i denote the ith prime number. By making a comparison with the harmonic series, show that the sequence of products

$$\pi_n = \left(1 - \frac{1}{p_1}\right)^{-1} \left(1 - \frac{1}{p_2}\right)^{-1} \cdots \left(1 - \frac{1}{p_n}\right)^{-1}$$

diverges by comparison with the harmonic series. [*Hint: you may assume that every integer greater than 1 can be uniquely written as a product of prime numbers.*]

(ii) Use power series to show that

$$-\log\left(1 - \frac{1}{x}\right) < \frac{2}{x} \qquad \text{for } \; x \geqslant 2.$$

(iii) Deduce that $\sum 1/p$ diverges, where the sum is taken over all primes p. (This result was first proved by Euler in 1737.)

Extension Exercises (Optional)

7. Use the binomial theorem for a general exponent to show that

$$(1-z)^{-1/2} = \sum_{k=0}^{\infty} \binom{2k}{k} \frac{z^k}{2^{2k}},$$

for $|z| < 1$. Carefully evaluate the sum

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2^{6k}} \binom{4k}{2k}.$$

8. Let $\omega = e^{2\pi i/3}$. Show that $1 + \omega + \omega^2 = 0$. Find the power series for $e^z + e^{\omega z} + e^{\omega^2 z}$. Evaluate

$$\sum_{n=0}^{\infty} \frac{8^n}{(3n)!}, \qquad \sum_{n=0}^{\infty} \frac{27^n}{(3n+1)!}.$$

Express your answers in forms that are clearly real numbers.

9. The *Bernoulli numbers* B_n are defined for $n \geqslant 0$ by the generating function

$$\sum_{n=0}^{\infty} \frac{B_n x^n}{n!} = \frac{x}{e^x - 1}.$$

(i) Show that $B_0 = 1, B_1 = \frac{-1}{2}, B_2 = \frac{1}{6}$.
(ii) Show that $B_k = 0$ for odd $k > 1$.
(iii) Show that

$$\cot x = \sum_{n=0}^{\infty} (-1)^n \frac{2^{2n} B_{2n} x^{2n-1}}{(2n)!}$$

(iv) Deduce that

$$\tan x = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2^{2n} \left(2^{2n} - 1\right) B_{2n} x^{2n-1}}{(2n)!}.$$

What is the radius of convergence of this power series?

10. The generating function for the *Legendre polynomials* $P_n(x)$, where $n \geqslant 0$, is

$$\frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n.$$

(i) Determine $P_0(x)$, $P_1(x)$ and $P_2(x)$.
(ii) Show that $P_n(1) = 1$ for all n.
(iii) Show that $P_n(x)$ has degree n.
(iv) Show that

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x).$$

(v) Show that

$$(1 - x^2)P_n''(x) - 2xP_n'(x) + n(n+1)P_n(x) = 0.$$

11. Let l^2 denote the space of real sequences $(x_n)_0^\infty$ such that $\sum {x_n}^2$ converges and l^∞ denote the space of bounded real sequences. We define

$$\begin{aligned} \langle (x_n), (y_n) \rangle &= \sum_0^\infty x_n y_n \qquad \text{for } (x_n), (y_n) \in l^2. \\ \|(x_n)\| &= \sup_{n \geqslant 0} |x_n| \qquad \text{for } (x_n) \in l^\infty. \end{aligned}$$

(i) Show that $\langle \quad , \quad \rangle$ is a well-defined inner product on l^2.
(ii) Show that $\| \quad \|$ does not satisfy the *parallelogram law*

$$\|(x_n + y_n)\|^2 + \|(x_n - y_n)\|^2 = 2\|(x_n)\|^2 + 2\|(y_n)\|^2 .$$

Deduce that the norm $\| \quad \|$ is not induced by any inner product on l^∞.
(iii) Find a bounded sequence in l^∞ which does not have a convergent subsequence.
(iv) Show that a Cauchy sequence in l^∞ converges.